

Notes: Panousi & Papanikolaou (JF, 2012)

Investment, Idiosyncratic Risk, and Ownership

Contents

1	Big picture	2
2	Incremental contribution	2
3	Data and measurement	2
3.1	Investment and risk	2
3.2	Ownership, compensation, and monitoring	2
4	Models and empirical specifications	3
4.1	Model	3
4.2	Baseline regression	3
4.3	Ownership interaction	3
5	Identification and empirical design	3
6	Tables and figures	3
6.1	Table I: Idiosyncratic Risk and Investment + Table II: Summary Statistics by Insider Ownership	3
6.2	Table III: Effect of Insider Ownership + Table IV: Option Exposure to Volatility	4
6.3	Table V: Institutional Ownership + Table VI: Endogeneity and Measurement Error	5
6.4	Table VII: Financial Constraints + Table VIII: Irreversibility and Competition	6
6.5	Figure 1: Idiosyncratic Volatility and Investment + Figure A.1: Optimal Ownership and Elasticity of Capital to Risk	7
7	Result map	8
8	Short summary	8
9	Conclusion	9
9.1	Core conclusion	9
9.2	Economic intuition	9
9.3	Implications	9
9.4	Limitations	9
9.5	Takeaway	9

1 Big picture

- **Question.** Does firm-specific risk affect corporate investment when managers are undiversified and risk averse?
- **Frictionless benchmark.** Diversified shareholders should care only about systematic risk, not idiosyncratic risk.
- **Managerial wedge.** Managers often hold firm shares and cannot hedge their own-company exposure, so idiosyncratic risk enters their objective.
- **Main prediction.** Investment falls with idiosyncratic risk, and the slope is more negative for firms with higher insider ownership.
- **Main result.** The investment-idiosyncratic-risk relation is negative, robust, and stronger in high-insider-ownership firms.
- **Mitigators.** Option convexity and institutional ownership weaken the effect, consistent with convex payoffs and monitoring.

2 Incremental contribution

- The paper shifts the investment-uncertainty literature from total uncertainty and real options toward **managerial exposure to idiosyncratic risk**.
- It connects executive ownership and compensation to real investment, showing a cost of high-powered incentives.
- It separates idiosyncratic volatility from systematic volatility and shows that the former is central to the ownership mechanism.
- It combines ownership interactions, option convexity, institutional monitoring, measurement-error corrections, and a customer-concentration instrument.
- It links firm-specific uncertainty to aggregate investment dynamics through managers who are not diversified.

3 Data and measurement

3.1 Investment and risk

Investment is capital expenditures scaled by lagged replacement capital, I_t/K_{t-1} . Idiosyncratic risk is estimated from weekly stock-return residuals after removing market and industry returns:

$$R_{i\tau} = a_{1i} + a_{2i}F_{i\tau} + \varepsilon_{i\tau}, \quad \log(\sigma_{it}) = \log\left(\sum_{\tau \in t} \varepsilon_{i\tau}^2\right).$$

Detailed interpretation. This isolates firm-specific risk that managers cannot diversify away.

3.2 Ownership, compensation, and monitoring

Insider ownership comes from Forms 3, 4, and 5. Option convexity is measured with Execucomp option vegas and deltas. Institutional ownership comes from 13F filings.

Detailed interpretation. Shares expose managers to firm-specific downside risk, options add convexity, and institutions may monitor investment decisions.

4 Models and empirical specifications

4.1 Model

$$\begin{aligned} y &= X\sqrt{K} + e, & U_0 &= u(c_0) - v(e) + \beta E_0 u(c_1), & u(c) &= -e^{-Ac}. \\ c_0 &= \lambda(C - K) - B + T, & c_1 &= \lambda(X\sqrt{K} + e) + RB. \\ K^* &= \left(\frac{\mu}{2R + \lambda A \sigma^2} \right)^2, & \frac{\partial \log K}{\partial \log \sigma^2} &= -\frac{\lambda A \sigma^2}{R + \frac{1}{2} \lambda A \sigma^2}. \end{aligned}$$

Detailed interpretation. Investment declines with idiosyncratic risk, and the elasticity is more negative when insider ownership or risk aversion is higher.

4.2 Baseline regression

$$\frac{I_{it}}{K_{i,t-1}} = \alpha_i + \delta_t + \beta \log(\sigma_{i,t-1}) + \Gamma Z_{i,t-1} + \varepsilon_{it}.$$

Detailed interpretation. The coefficient β measures the average relation between lagged idiosyncratic volatility and investment.

4.3 Ownership interaction

$$\frac{I}{K} = \alpha_i + \delta_t + \sum_q \beta_q (INSD_q \times \log \sigma_{i,t-1}) + \Gamma Z + \varepsilon.$$

Detailed interpretation. The key test is whether $\beta_{high} - \beta_{low} < 0$.

5 Identification and empirical design

The design combines firm and time fixed effects, standard investment controls, systematic-volatility controls, alternative growth-opportunity measures, Erickson-Whited measurement-error estimators, a customer-concentration instrument for idiosyncratic risk, and matched/grouped comparisons by size, constraints, irreversibility, competition, and market power. The mechanism is further tested with option convexity, institutional ownership, and the 2008–2009 crisis.

6 Tables and figures

6.1 Table I: Idiosyncratic Risk and Investment + Table II: Summary Statistics by Insider Ownership

Read: Table I establishes the negative investment-idiosyncratic-risk relation. Table II shows that high-insider firms differ from low-insider firms, motivating controls.

Detailed interpretation. Table I gives the main empirical fact; Table II identifies the selection problem.

Economic message. Idiosyncratic risk matters, but ownership comparisons require careful controls.

Table I
Idiosyncratic Risk and Investment

Table I reports estimation results of equation (11), where the dependent variable is the investment rate (I_t/K_{t-1}). Our baseline measure of risk, $\sigma_{i,t-1}$, is constructed from a regression of weekly firm-level returns on the CRSP VW index and the corresponding industry portfolio. Additional regressors include lagged values of Tobin's Q (Q_{t-1}) defined as in Fazzari, Hubbard, and Petersen (1988); operating cash flows (CF_{t-1}/K_{t-2}) defined as the ratio of operating income (item 18) to the replacement cost of capital, computed as in Salinger and Summers (1983); the firm's size ($\log(\tilde{K}_{t-1})$) defined as the log value of its replacement cost of capital, scaled by average capital across all firms; the firm's stock return (R_{t-1}); leverage (E_{t-1}/A_{t-1}) defined as the ratio of book equity (item 216) to book assets (item 6); and systematic volatility ($\log(\sigma_{t-1}^{syst})$) defined as the (log of the) square root of the difference between the firm's total variance and its idiosyncratic variance. Item refers to Compustat items. The sample period is 1970 to 2005. Here, F denotes firm fixed effects, T denotes time fixed effects, and $I \times T$ denotes industry-time fixed effects. The standard errors are clustered at the firm-level, and t -statistics are reported in parentheses.

I_t/K_{t-1}	No Controls	BENCH	$IND \times T$	Syst
$\log(\sigma_{i,t-1})$	-0.0346 (-13.78)	-0.0196 (-8.44)	-0.0197 (-8.32)	-0.0244 (-9.94)
$\log(\sigma_{i,t-1}^{syst})$				0.0063 (5.51)
$\log(Q_{t-1})$		0.0699 (45.05)	0.0692 (43.41)	0.0690 (44.31)
CF_{t-1}/K_{t-2}		0.0222 (9.91)	0.0218 (9.88)	0.0220 (9.84)
$\log(\tilde{K}_{t-1})$		-0.1179 (-31.80)	-0.1243 (-31.67)	-0.1188 (-32.09)
R_{t-1}		0.0146 (8.66)	0.0131 (7.47)	0.0148 (8.75)
$\log(E_{t-1}/A_{t-1})$		0.0352 (14.13)	0.0346 (13.82)	0.0346 (13.96)
Observations	104,646	104,646	104,646	104,619
R^2	0.399	0.563	0.569	0.563
Fixed effects	F	F, T	$F, I \times T$	F, T
Estimation Method	OLS	OLS	OLS	OLS

(a) Table I: Idiosyncratic Risk and Investment

Table II

Summary Statistics: Five Portfolios Sorted on Insider Ownership

This table shows time-series averages of characteristics of the five portfolios sorted by levels of insider ownership. All characteristics, except for Tobin's Q , are in percent. We scale firm market capitalization and book assets at time t by the average across all firms at time t . Every year, we sort firms into quintiles based on the fraction of shares outstanding owned by company officers. See the Appendix for details on the construction of our insider ownership measure. Accounting data are from Compustat: book debt is item 9 + item 34; cash is item 1; dividends is item 19 + item 21; cash flows is item 14 + item 18; and sales is item 12. Market capitalization is the December market capitalization of equity from CRSP. Remaining variables are defined in the notes to Table I. We report the time-series average of median portfolio characteristics. The sample period is 1987 to 2005.

Level of insider ownership	Low	2	3	4	High
Insider ownership	0.06	0.32	1.01	3.40	17.43
Market capitalization (scaled by mean)	20.84	17.23	11.01	6.61	3.51
Book assets (scaled by mean)	27.38	21.04	13.13	7.44	4.38
Tobin's Q	1.64	1.80	1.93	2.27	2.15
Cash flows to capital	12.27	14.41	15.17	15.98	16.05
Investment to capital	8.76	9.41	10.57	12.18	12.47
Idiosyncratic volatility	35.37	36.65	40.74	44.62	49.11

(b) Table II: Summary Statistics by Insider Ownership

6.2 Table III: Effect of Insider Ownership + Table IV: Option Exposure to Volatility

Read: Table III shows that high-insider firms have a more negative investment-risk sensitivity. Table IV shows that option convexity weakens this sensitivity.

Detailed interpretation. The two tables jointly support a managerial-risk-aversion channel.

Economic message. Shares can induce underinvestment under idiosyncratic risk; options can mitigate it.

Table III
Effect of Insider Ownership

This table reports estimation results of equation (11), interacting our idiosyncratic volatility measure ($\sigma_{i,t-1}$) with insider ownership quintile dummies (see Appendix for details). Here, $INSD_L$ and $INSD_H$ indicate the dummies for the first and the fifth insider ownership quintile, respectively. The third and fourth columns adjust insider ownership for option holdings, whenever available, otherwise option holdings are assumed to be zero. Depending on the specification, we include firm (F) and year (T) fixed effects. The set of controls includes Tobin's Q , operating cash flows, firm size, the firm's stock return, and leverage (see notes to Table I for details). We interact time and firm fixed effects with quintile dummies. The standard errors are clustered at the firm level, and t -statistics are reported in parentheses.

I_i/K_{i-1}	(1)	(2)	(3)	(4)
$INSD_L \times \log(\sigma_{i,t-1})$	-0.0089 (-1.10)	-0.0049 (-0.62)	-0.0012 (-0.12)	-0.0046 (-0.51)
$INSD_2 \times \log(\sigma_{i,t-1})$	-0.0155 (-2.02)	-0.0045 (-0.62)	-0.0086 (-0.95)	-0.0019 (-0.22)
$INSD_3 \times \log(\sigma_{i,t-1})$	-0.0245 (-2.90)	-0.0070 (-0.83)	-0.0201 (-1.95)	-0.0109 (-1.06)
$INSD_4 \times \log(\sigma_{i,t-1})$	-0.0400 (-4.13)	-0.0161 (-1.72)	-0.0312 (-2.61)	-0.0190 (-1.65)
$INSD_H \times \log(\sigma_{i,t-1})$	-0.0413 (-3.52)	-0.0305 (-2.93)	-0.0408 (-2.81)	-0.0331 (-2.60)
$INSD_H - INSD_L$	-0.0324	-0.0255	-0.0396	-0.0285
p -value	0.022	0.049	0.023	0.066
Observations	41,206	41,206	32,616	32,616
Fixed effects	F	F, T	F	F, T
Controls	N	Y	N	Y
Option-adjusted ownership	N	N	Y	Y

(a) Table III: Effect of Insider Ownership

Table IV
Effect of Insider's Option Exposure to Volatility

This table reports estimation results of equation (11) separately for firms with different levels of insider ownership and convexity of executive compensation. See the Appendix for details on the construction of our compensation convexity and insider ownership measures. Every year, we first sort firms into terciles based on ownership by insiders (High, Medium, Low) and then sort into terciles based on the sensitivity of compensation to volatility (High, Medium, Low). We report results for the four corner portfolios. The sample period is 1992 to 2005. Depending on the specification, we include firm (F) and year (T) fixed effects. The set of controls includes Tobin's Q , operating cash flows, firm size, the firm's stock return, and leverage (see notes to Table I for details). We interact time and firm fixed effects with quintile dummies. The standard errors are clustered at the firm level, and t -statistics are reported in parentheses. We report the Chow test p -value for the null hypothesis that the idiosyncratic volatility, $\sigma_{i,t-1}$, coefficients on the high and low vega groups are the same.

I_i/K	(1)	(2)	(3)	(4)
$INSD_L \times VEGA_L \times \log(\sigma_{i,t-1})$	-0.0221 (-2.10)	-0.0172 (-1.87)	-0.0223 (-2.10)	-0.0175 (-1.90)
$INSD_L \times VEGA_H \times \log(\sigma_{i,t-1})$	0.0112 (1.72)	0.0127 (1.79)	0.0112 (1.73)	0.0108 (1.57)
$VEGA_H - VEGA_L$	0.0333	0.0299	0.0335	0.0283
p -value	0.007	0.009	0.007	0.013
$INSD_H \times VEGA_L \times \log(\sigma_{i,t-1})$	-0.0315 (-2.79)	-0.0286 (-2.83)	-0.0320 (-2.81)	-0.0296 (-2.91)
$INSD_H \times VEGA_H \times \log(\sigma_{i,t-1})$	0.0152 (1.32)	-0.0001 (-0.01)	0.0148 (1.28)	-0.0007 (-0.07)
$VEGA_H - VEGA_L$	0.0467	0.0285	0.0468	0.0289
p -value	0.004	0.054	0.004	0.052
Total observations	12,708	12,708	12,708	12,708
Fixed effects	F	F, T	F	F, T
Controls	N	Y	N	Y
Option-adjusted ownership	N	N	Y	Y

Within each group, we then sort firms into three equal-sized groups based on the level of insider ownership. We estimate equation (11) separately for firms in each of the 3×3 groups and we report results for the four corners in Table V. Columns 1 and 2 show results using our benchmark (not adjusted for options) measure of insider ownership, while columns 3 and 4 show results for our options-adjusted measure.

(b) Table IV: Option Exposure to Volatility

6.3 Table V: Institutional Ownership + Table VI: Endogeneity and Measurement Error

Read: Table V shows weaker ownership-risk sensitivity when institutional ownership is high. Table VI shows robustness to endogeneity and measurement error.

Detailed interpretation. Institutional investors appear to mitigate the underinvestment distortion; robustness tests reduce concerns that risk proxies omitted growth opportunities.

Economic message. Monitoring and better identification both point toward the same mechanism.

Table V
Effect of Insider Ownership by Level of Institutional Ownership

This table reports estimation results of equation (11) separately for firms with different levels of institutional ownership and insider ownership. See the Appendix for details on the construction of our insider and institutional ownership measures. Every year, we first sort firms into three equal-sized groups based on institutional ownership (High, Medium, Low) and then sort into three equal-sized groups based on ownership by insiders (High, Medium, Low). We report results for the four corner portfolios. The sample period is 1992 to 2005. Depending on the specification, we include firm (F) and year (T) fixed effects. The set of controls includes Tobin's Q , operating cash flows, firm size, the firm's stock return, and leverage (see notes to Table I for details). We interact time and firm fixed effects with quintile dummies. The standard errors are clustered at the firm level, and t -statistics are reported in parentheses. We report the Chow test p -value for the null hypothesis that the idiosyncratic volatility, $\sigma_{i,t-1}$, coefficients on the high and low insider ownership groups are the same.

I/K	(1)	(2)	(3)	(4)
$INST_L \times INSD_L \times \log(\sigma_{i,t-1})$	-0.0110 (-1.12)	0.0042 (0.45)	-0.0116 (-1.03)	0.0050 (0.45)
$INST_L \times INSD_H \times \log(\sigma_{i,t-1})$	-0.0627 (-5.21)	-0.0359 (-3.40)	-0.0740 (-5.04)	-0.0453 (-3.68)
$INSD_H - INSD_L$	-0.0517 0.001	-0.0401 0.004	-0.0624 0.001	-0.0503 0.002
$INST_H \times INSD_L \times \log(\sigma_{i,t-1})$	-0.0077 (-1.34)	0.0005 (0.08)	0.0005 (0.08)	-0.0009 (-0.11)
$INST_H \times INSD_H \times \log(\sigma_{i,t-1})$	-0.0098 (-0.97)	0.0041 (0.38)	0.0047 (0.41)	0.0001 (0.00)
$INSD_H - INSD_L$	-0.0021 0.857	0.0036 0.775	0.0042 0.752	0.0010 0.952
Total observations	40,705	40,705	32,230	32,230
Fixed effects	F N	F, T V	F N	F, T V

(a) Table V: Institutional Ownership

Table VI
Endogeneity and Measurement Error

This table reports estimation results of a modified version of equation (11), where the dependent variable is the investment rate (I_t/K_{t-1}). Here, Q^{bes} is constructed from analysts' forecasts as in Cummins, Hassett, and Oliner (2006). Next, β^{inc} refers to the firm's univariate beta with a portfolio long the capital-producing sector and short the consumption-producing sector (IMC); see Kogan and Papanikolaou (2010a,b) for more details. In columns 1 to 3, the set of additional controls includes operating cash flows, firm size, the firm's stock return, and leverage (see notes to Table I for details). In columns 4 and 5, we include the lagged investment rate and we replace firm size with the firm's total sales (scaled by industry sales at the three-digit SIC level) as controls. Depending on the specification, we include firm (F), industry (I), and/or year (T) fixed effects. Column 5 presents results using the Erickson and Whited (2000) estimator. We report the p -value of a joint test for the null of no identification across all time periods using the Bonferroni method and the r^2 statistic. Columns 6 and 7 present estimation results using instrumental variables, where we instrument for idiosyncratic risk ($\sigma_{i,t-1}$) with the concentration of a firm's customer base, $H_{i,t}$, constructed using customer data from the Compustat segment files (see text for details). We report the value of the Kleibergen-Paap rk Wald F -test of the null of weak instruments. The critical value assuming 10% maximal IV size equals 16.38 (Stock and Yogo (2005)). In columns 1 to 3 and 6 to 7, standard errors are clustered at the firm level. In columns 4 and 5, standard errors are computed using the Fama and McBeth (1973) procedure.

I_t/K_{t-1}	Q^{bes}	β^{inc}	$Q \& \beta^{inc} \& Q^{bes}$	OLS	EW_2	IV
$\log(\sigma_{i,t-1})$	-0.0175 (-4.78)	-0.0464 (-15.60)	-0.0090 (-2.09)	-0.0366 (-11.95)	-0.0323 (-6.18)	-0.1167 (-7.00)
Q^{bes}_{t-1}	0.0435 (16.83)		0.0248 (6.00)			-0.1483 (-6.88)
β^{inc}_{t-1}		0.0074 (7.58)	0.0016 (0.93)			
Q_{t-1}				0.0032 (9.48)	0.0079 (2.20)	
$\log(Q_{t-1})$			0.0512 (14.81)			0.0569 (45.48)
Observations	33,628	77,001	24,904	104,646	104,646	91,157
Bonferroni test (p -value)					0.001	
r^2 statistic					0.465	
Kleibergen-Paap rk Wald F statistic						299.3
Fixed effects	F, T	F, T	F, T	F, T	F, T	I, T
Controls	Y	Y	Y	Y	Y	Y
Estimation method	OLS	OLS	OLS	OLS	GMM	IV

(b) Table VI: Endogeneity and Measurement Error

6.4 Table VII: Financial Constraints + Table VIII: Irreversibility and Competition

Read: Table VII controls for financial constraints. Table VIII controls for irreversibility, market power, and industry competition.

Detailed interpretation. The ownership-risk channel is not explained away by financing constraints or real-option/product-market channels.

Economic message. The mechanism survives comparisons among more similar firms.

Table VII

Effect of Insider Ownership, Controlling for Financial Constraints

This table reports estimation results of equation (11) separately for firms with different levels of insider ownership, controlling for firm size and degree of financial constraints. Every year, we sort firms into 5×5 groups: based on firm size or the degree of financial constraints and then based on the level of insider ownership. We use three measures of financial constraints: the Whited and Wu (2006) index (WW), the firm's S&P credit rating (RT), and the book value of assets (A). We group firms with a similar credit rating as follows: group 1 contains firms rated AA- or better, group 2 contains firms rated A, group 3 contains firms rated BBB, group 4 contains firms rated BB+ or lower, and group 5 contains unrated firms. The sample period is 1987 to 2005. Depending on the specification, we include firm (F) and year (T) fixed effects. The set of controls includes Tobin's Q , operating cash flows, firm size, the firm's stock return, and leverage (see notes to Table I for details). We interact time and firm fixed effects with quintile dummies. Standard errors are clustered at the firm level, and t -statistics are reported in parentheses. We report the difference in coefficients on idiosyncratic risk across quintiles 1 and 5 and the Chow-test p -value for the null hypothesis that the coefficients are equal.

I/K	(1)	(2)	(3)	(4)	(5)	(6)
$INSD_L \times \log(\sigma_{i,t-1})$	-0.0070 (-0.77)	0.0050 (0.57)	-0.0046 (-0.47)	-0.0009 (-0.11)	-0.0055 (-0.61)	0.0035 (0.40)
$INSD_2 \times \log(\sigma_{i,t-1})$	-0.0179 (-1.94)	-0.0091 (-1.03)	-0.0199 (-2.24)	-0.0111 (-1.31)	-0.0242 (-2.41)	-0.0140 (-1.54)
$INSD_3 \times \log(\sigma_{i,t-1})$	-0.0237 (-2.29)	-0.0146 (-1.55)	-0.0185 (-1.79)	-0.0073 (-0.71)	-0.0286 (-2.83)	-0.0170 (-1.85)
$INSD_4 \times \log(\sigma_{i,t-1})$	-0.0372 (-3.98)	-0.0028 (-0.31)	-0.0353 (-3.78)	-0.0148 (-1.61)	-0.0412 (-4.36)	-0.0123 (-1.28)
$INSD_H \times \log(\sigma_{i,t-1})$	-0.0285 (-2.53)	-0.0275 (-2.66)	-0.0322 (-3.05)	-0.0241 (-2.49)	-0.0311 (-2.70)	-0.0280 (-2.63)
$INSD_H - INSD_L$	-0.0215	-0.0325	-0.0277	-0.0232	-0.0256	-0.0315
p -value	0.059	0.002	0.014	0.023	0.026	0.004
Observations	40,985	40,985	40,773	40,773	41,206	41,206
Financial constraints	WW	WW	RT	RT	A	A
Fixed effects	F	F, T	F	F, T	F	F, T
Controls	N	Y	N	Y	N	Y

(a) Table VII: Financial Constraints

Table VIII

Effect of Insider Ownership, Controlling for Investment Irreversibility and Imperfect Competition

This table reports estimation results of equation (11) separately for firms with different levels of insider ownership, controlling for the degree of investment irreversibility and industry competition or firm market power. Every year, we sort firms into $5 \times 5 \times 3$ groups based on the degree of investment irreversibility, then on the degree of firm market power (top panel) or industry competition (bottom panel), and then on the level of insider ownership. We use five measures of investment irreversibility: age-adjusted price discount between new and old capital (P), using data from Ramey and Shapiro (2001); the mean depreciation rate of the industry, (δ); the firm's beta with the corresponding industry portfolio, (β^{ind}); the ratio of new to used investment at the industry level from the Census Bureau (U); and the average ratio of sales of property, plant, and equipment to capital for the industry during the 1970 to 1990 period, (S). We use the Herfindahl index as a measure of industry competition and the ratio of firm sales to total industry sales as a measure of firm market power. Industries are defined at the three-digit SIC code. Sample period is 1987 to 2005.

$I/K, \lambda$	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
A: Controlling for degree of investment irreversibility and firm market power										
$INSD_L \times \log(\sigma_{i,t-1})$	0.0024 (0.10)	0.0228 (0.80)	-0.0127 (-1.91)	-0.0044 (-0.72)	-0.0126 (-2.02)	-0.0012 (-0.20)	-0.0134 (-2.23)	-0.0043 (-0.72)	-0.0129 (-2.06)	-0.0036 (-0.58)
$INSD_2 \times \log(\sigma_{i,t-1})$	-0.0231 (-2.75)	-0.0008 (-0.27)	-0.0256 (-3.94)	-0.0134 (-2.00)	-0.0259 (-3.92)	-0.0114 (-1.68)	-0.0276 (-4.19)	-0.0149 (-2.67)	-0.0219 (-3.17)	-0.0083 (-1.28)
$INSD_3 \times \log(\sigma_{i,t-1})$	-0.0016 (-0.32)	-0.0648 (-1.94)	-0.0377 (-4.89)	-0.0214 (-3.03)	-0.0338 (-4.41)	-0.0206 (-2.96)	-0.0407 (-5.31)	-0.0258 (-3.72)	-0.0454 (-6.11)	-0.0297 (-4.34)
$INSD_H - INSD_L$	-0.0640	-0.0674	-0.0256	-0.0171	-0.0212	-0.0194	-0.0273	-0.0215	-0.0325	-0.0361
p -value	0.048	0.036	0.013	0.062	0.028	0.033	0.004	0.015	0.001	0.004
Observations	2,184	2,184	41,179	41,179	40,454	40,454	40,661	40,661	40,828	40,828
B: Controlling for degree of investment irreversibility and industry competition										
$INSD_L \times \log(\sigma_{i,t-1})$	0.0238 (1.15)	0.0291 (1.53)	-0.0093 (-1.67)	-0.0018 (-0.34)	-0.0074 (-1.27)	0.0012 (0.21)	-0.0124 (-2.26)	-0.0007 (-0.14)	-0.0091 (-1.56)	0.0001 (0.03)
$INSD_2 \times \log(\sigma_{i,t-1})$	-0.0481 (-1.42)	-0.0047 (-0.13)	-0.0281 (-4.20)	-0.0100 (-1.59)	-0.0281 (-4.43)	-0.0076 (-1.19)	-0.0335 (-5.17)	-0.0130 (-1.80)	-0.0284 (-4.33)	-0.0105 (-1.60)
$INSD_3 \times \log(\sigma_{i,t-1})$	-0.0630 (-1.74)	-0.0536 (-1.45)	-0.0438 (-5.46)	-0.0275 (-3.59)	-0.0404 (-4.97)	-0.0255 (-3.36)	-0.0417 (-5.10)	-0.0274 (-3.64)	-0.0442 (-5.47)	-0.0293 (-3.91)
$INSD_H - INSD_L$	-0.0868	-0.0827	-0.0445	-0.0257	-0.0331	-0.0267	-0.0289	-0.0286	-0.0332	-0.0295
p -value	0.046	0.046	0.000	0.005	0.001	0.003	0.004	0.004	0.000	0.001
Observations	2,184	2,184	41,028	41,028	40,395	40,395	40,517	40,517	40,682	40,682
Investment irreversibility measure	P	P	δ	δ	β^{ind}	β^{ind}	U	U	S	S
Fixed effects	F	F, T	F	F, T	F	F, T	F	F, T	F	F, T
Controls	N	Y	N	Y	N	Y	N	Y	N	Y

(b) Table VIII: Irreversibility and Competition

6.5 Figure 1: Idiosyncratic Volatility and Investment + Figure A.1: Optimal Ownership and Elasticity of Capital to Risk

Read: Figure 1 visualizes the crisis test and the time-series relation between aggregate idiosyncratic volatility and the investment gap by insider ownership. Figure A.1 shows the model-implied optimal ownership and elasticity of capital to risk.

Detailed interpretation. The crisis increased uncertainty broadly, but investment fell most among high-insider firms; the model explains why optimal incentive contracts do not fully remove this distortion.

Economic message. The incentives-versus-insurance tradeoff has real investment consequences.

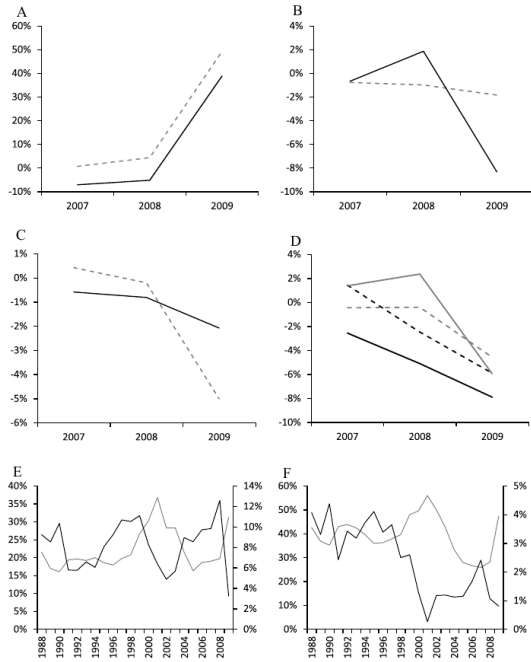


Figure 1. Idiosyncratic Volatility and Investment. (A) Change in log idiosyncratic volatility (relative to 2006) for firms with high (solid line) and low (dotted line) insider ownership. (B) Change in portfolio investment rates (relative to 2006) for firms with high (solid line) and low (dotted line) insider ownership firms. (C) Change in portfolio investment rates (relative to 2006) for firms with high (solid line) and low (dotted line) compensation convexity. (D) Change in portfolio investment rates (relative to 2006) for firms with high (solid line) and low (dotted line) insider ownership. (E) Black line is difference in average (weighted by capital stock) portfolio investment rates between high and low insider ownership firms (right axis); grey line is average (weighted by capital stock) idiosyncratic volatility across firms (left axis). (F) Black line is difference in median portfolio investment rates between high and low insider ownership firms (right axis); grey line is median

(a) Figure 1: Idiosyncratic Volatility and Investment

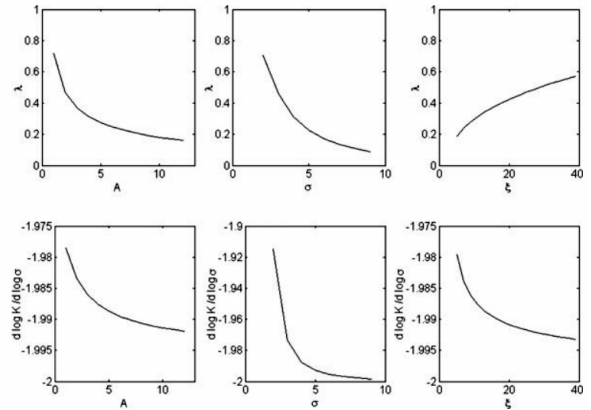


Figure A.1. Optimal ownership and elasticity of capital to risk. This figure plots the solution of the model in terms of the optimal share of insider ownership, λ^* , as a function of three

(b) Figure A.1: Optimal Ownership and Elasticity of Capital to Risk

7 Result map

- Baseline: idiosyncratic volatility is negatively related to investment.
- Ownership: high-insider firms have more negative investment-risk sensitivity.
- Options: compensation convexity reduces the negative slope.
- Institutions: institutional ownership reduces the insider-ownership mechanism.
- Robustness: the results survive growth controls, measurement-error corrections, IV estimates, financial-constraint controls, and product-market controls.
- Crisis: high-insider firms reduce investment more during the 2008–2009 uncertainty shock.

8 Short summary

The paper shows that firm-specific risk affects investment when managers are undiversified. The model predicts that investment falls with idiosyncratic risk and that this fall is larger with more insider ownership. The data support both predictions, and the mechanism is strengthened by evidence on option convexity, institutional monitoring, customer concentration, and the financial crisis.

9 Conclusion

9.1 Core conclusion

Idiosyncratic risk can affect corporate investment because managers, unlike diversified shareholders, personally bear firm-specific risk.

9.2 Economic intuition

Equity ownership gives managers incentives, but it also exposes them to firm-specific risk. When uncertainty rises, risk-averse managers privately prefer lower investment than diversified shareholders would choose.

9.3 Implications

The paper implies that shares, options, and monitoring are not interchangeable governance tools. Shares strengthen incentives but can induce risk-averse underinvestment. Options and institutional monitoring can mitigate the distortion.

9.4 Limitations

Managerial risk aversion is not directly observed, and ownership and compensation are endogenous. The paper addresses these concerns with many robustness tests, but the design is not a randomized experiment.

9.5 Takeaway

High-powered incentives have a real cost: they can turn diversifiable risk into a determinant of corporate investment.